

CONTACT

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San Francisco Bay Area

TECHNICAL SKILLS

Spark Hive Hadoop Databricks Delta
Lake Snowflake Redshift dbt
Airflow Dataswarm AWS Azure
GCP Terraform Kubernetes Python
SQL Scala LLM/RAG Pipelines Claude
Code AI Agents Prompt Eng.
LangChain GitHub Copilot MLflow
Diff. Privacy Unity Catalog Data Mesh

EDUCATION

Data Engineering Cert.
UC Santa Cruz, 2013–14
B.S. Electrical Engineering
Sardar Patel College of Eng.
Diploma, Electronics & Comms
Technical Board, Tamil Nadu

PUBLICATIONS

Cloud Computing
All About Big Data
Future Trends in BI

AI & THOUGHT LEADERSHIP

128 articles on paddyspeaks.com
Claude Code skill development
Vibe coding methodology
AI agent architecture writing

OPEN-SOURCE & COMMUNITY

Interview Studio — free Data & AI
Engineering prep
22 data-model deep-dives +
1,500+ questions
In-browser SQL / Python (WASM),
no backend

EARLIER CAREER

Data Engineer Consultant

[LinkedIn](#)

Dec 2017 – Apr 2018
GDPR compliance — PII encryption into
Dali storage (Hive, Python, Pig)

Data Engineer

[GREE International](#)

2014 – 2015
PII masking, Vertica→Redshift, 1,000+
table optimization

Data Architect

[Chegg Inc.](#)

Oct 2013 – Apr 2014

Technical Director

[Model N](#)

2011 – 2013
Life Sciences BI, cloud migration, ETL
modularization

Architect

[CallidusCloud](#)

2005 – 2011
Incentive comp analytics, BusinessObjects
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Paddy Iyer

DATA ENGINEERING LEADER • STRATEGIC DATA ARCHITECT • AI & PRIVACY ENGINEERING • AI INNOVATOR

35+ years building and transforming enterprise data platforms for global tech leaders across cloud/SaaS, fintech, gaming, and consumer platforms. Expert in cloud-native architectures, real-time analytics, privacy-first pipelines, and AI-augmented data operations. Proven builder of high-trust, high-performance data teams at petabyte scale. Published writer and thought leader on data architecture, AI agents, and the intersection of ancient wisdom with modern technology.

Modern Data Architecture Privacy-First Design Consent Architecture AI Agents & LLM Pipelines Prompt Engineering Data Governance (C1–C5) Petabyte-Scale Systems Global Team Leadership Data Storytelling

FEATURED EXPERIENCE

Data Engineer — Privacy & Consent Infrastructure

Present

Meta

- Delivered ~\$500K/year in recovered engineering capacity by sustaining the output of 3–4 engineers as a single engineer through AI-assisted development — 1,070 production changes landed at ~1.7/working day over 2.5 years, peaking at 262 in a single month
- Eliminated ~52,000 redundant data operations per year, removing an estimated 2–4 FTE of recurring manual toil (~\$500K–\$1M annually in engineering time, storage, and metadata overhead)
- Achieved this leverage at under \$1,000 in total LLM spend — roughly 500x return on inference cost — by optimizing on cost per landed change (~\$0.80) rather than cost per API call
- Cut LLM input tokens ~60% via cacheable prompt prefixes and routed simple transforms to zero-cost local engines, holding per-change cost flat as volume scaled
- Cut manual cross-namespace data imports 91% (156.8/day → 14.2/day), driving the final phase of a multi-year warehouse consolidation from 80% to 100% coverage
- Unblocked the last ~20% of warehouse data — the segment no prior effort could touch because it sat in the direct ads revenue path, where a single day of degradation carries seven-figure impact — and delivered it with zero revenue loss
- Reduced engineer ramp-up from 2 weeks to 2 days and per-asset cycle time 3–5x by building a reusable AI agent playbook, with every hard-block gate derived from a production postmortem
- Operated under a constraint where errors surface as revenue loss rather than test failures, which drove the guard-first design: proved the pre-deployment validation stack was structurally blind — permission checks passed for tables that did not exist — and built the system that closed the gap
- Collapsed an entire failure class into an offline-checkable invariant: 81% of migrations required a namespace change, not the name change the tooling assumed
- Owled incident response on revenue-critical infrastructure — 5 production incidents including 2 high-severity — resolved fix-forward with zero reverts and permission grants coordinated across 6+ teams

EARLIER CAREER

Lead Data Architect — Partner Data Engineering

VMware

Strategic architect for VMware's Partner Data Platform — unifying fragmented partner data into a single governed analytics layer

- Designed and built the Partner Data Platform from the ground up, unifying partner data from 10+ disparate sources into a single governed analytics layer — eliminating silos that had blocked cross-functional decision-making for years

DW Architect

Hewlett Packard

2001 - 2005

Enterprise DW + 8 datamarts; ETL 18→3 hrs; 40-50% sales lift via clickstream

- Implemented VMware's first structured data governance for partner data (classification, lineage, policy enforcement) plus a self-service data catalog — cutting tribal-knowledge reliance by 60% and ad-hoc request volume by ~40%, and achieving audit-ready status
- Pioneered an AI copilot for retrieval optimization, coding, and governance — one of VMware's earliest LLM-assisted data engineering deployments, reducing routine query development time by ~30%
- Achieved 40%+ Spark processing improvements via broadcast joins, caching, and skew management — bringing SLA-critical reports inside their delivery windows for the first time
- Engaged director/VP-level stakeholders to define a unified data vision, securing budget and headcount that grew the team from 3 to 8 engineers during the engagement

Data Engineer (Consultant) Oct 2015 - Dec 2021

Meta

Performance, query-engine migration and reliability across Meta's ads, integrity, commerce and marketing data

- Migrated 89 production pipelines from Hive to Spark and Presto over 11 months (~8/month sustained) — datelist, cube and funnel workloads across ads, integrity, commerce and marketing
- Benchmarked rather than assumed: on one workload Spark failed 40 times and ran 50+ minutes while Presto finished in under 30 — shipped Presto with documented evidence, avoiding a slower, failure-prone pipeline
- Cut a critical ads pipeline from 5 hours to under 1 hour by decomposing serial split tasks into 30 parallel threads, recovering ~4 hours of daily batch window on a revenue-reporting dependency
- Led privacy remediation across 1,200+ SMB 2.0, Customer Journey and BPO tables, and drove Salesforce ID deprecation across 1,000+ pipelines — eliminating an external vendor dependency
- Carried pipeline oncall across ~200 incidents with root-cause fixes rather than restarts; delivered ~50 privacy and compliance remediations and cut storage cost across ~60 retention engagements
- Recognized by 45 colleagues across 53 peer commendations for debugging, escalation and cross-team unblocking across ads, payments, commerce, marketplace and privacy